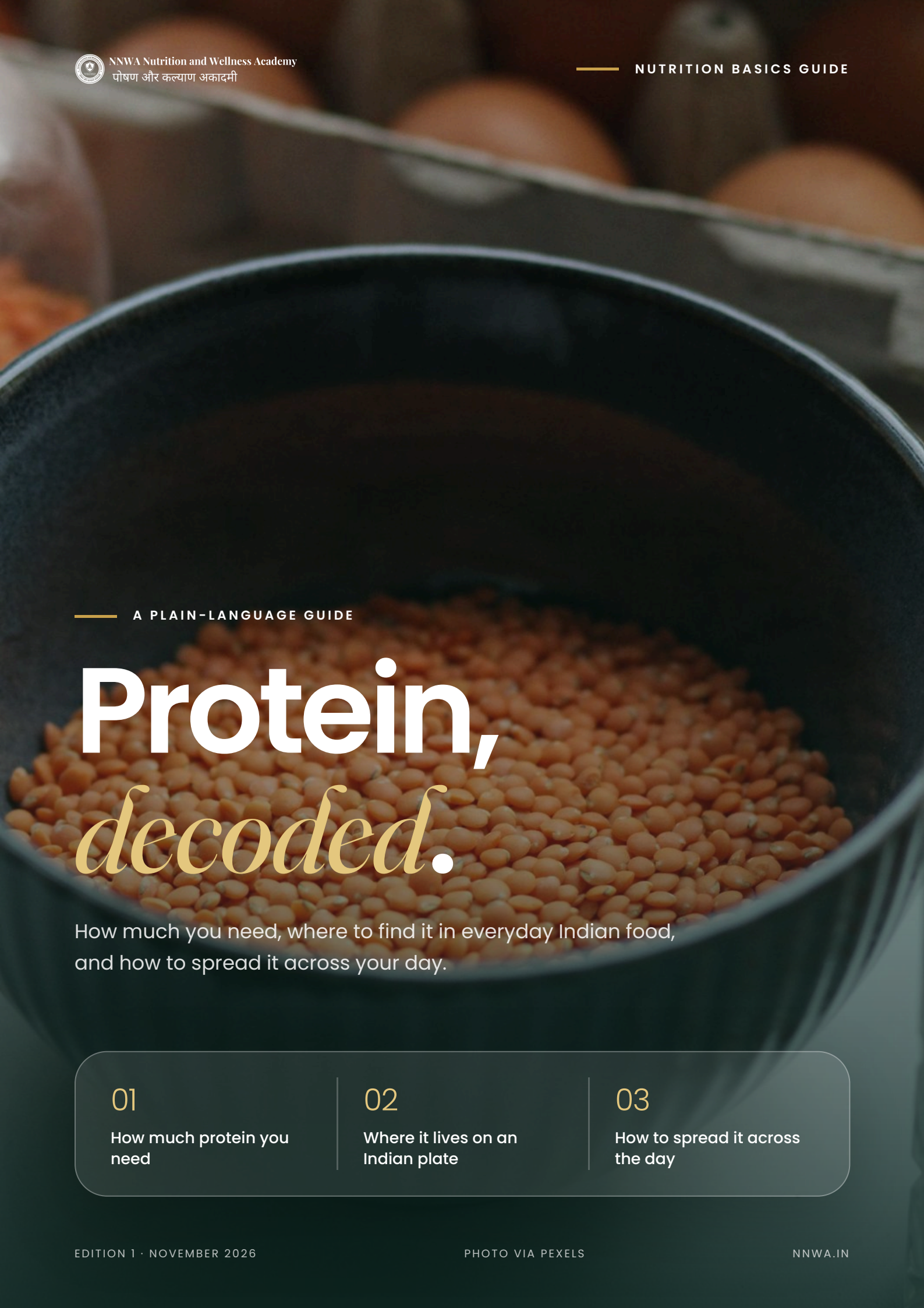


A PLAIN-LANGUAGE GUIDE

Protein, *decoded.*

How much you need, where to find it in everyday Indian food, and how to spread it across your day.

01
How much protein you need

02
Where it lives on an Indian plate

03
How to spread it across the day

INSIDE THIS GUIDE

Seven short *chapters*.

01 **What protein does** — 03
The jobs it does in your body, and why food has to supply some of it.

02 **How much you need** — 04
One number for healthy adults, and when that number changes.

03 **Where to find it** — 06
Approximate protein in common Indian servings, side by side.

04 **Pair it up** — 08
Why dal with rice works better than either one alone.

05 **Spread it across your day** — 09
An illustrative day of meals that adds up without a shake.

06 **Myths, checked** — 10
Five things people repeat about protein, against the evidence.

07 **Supplements, carefully** — 11
What Indian dietary guidance says, and what to check on a label.

HOW TO USE THIS GUIDE

Read it once end to end, then keep the checklist on page 12 on your fridge or phone. Every figure is listed with its source on page 13.

BEFORE YOU START

This is general information for healthy adults, not medical advice. If you are pregnant, breastfeeding, under 18, older, managing a medical condition or training heavily, speak to a qualified healthcare professional.

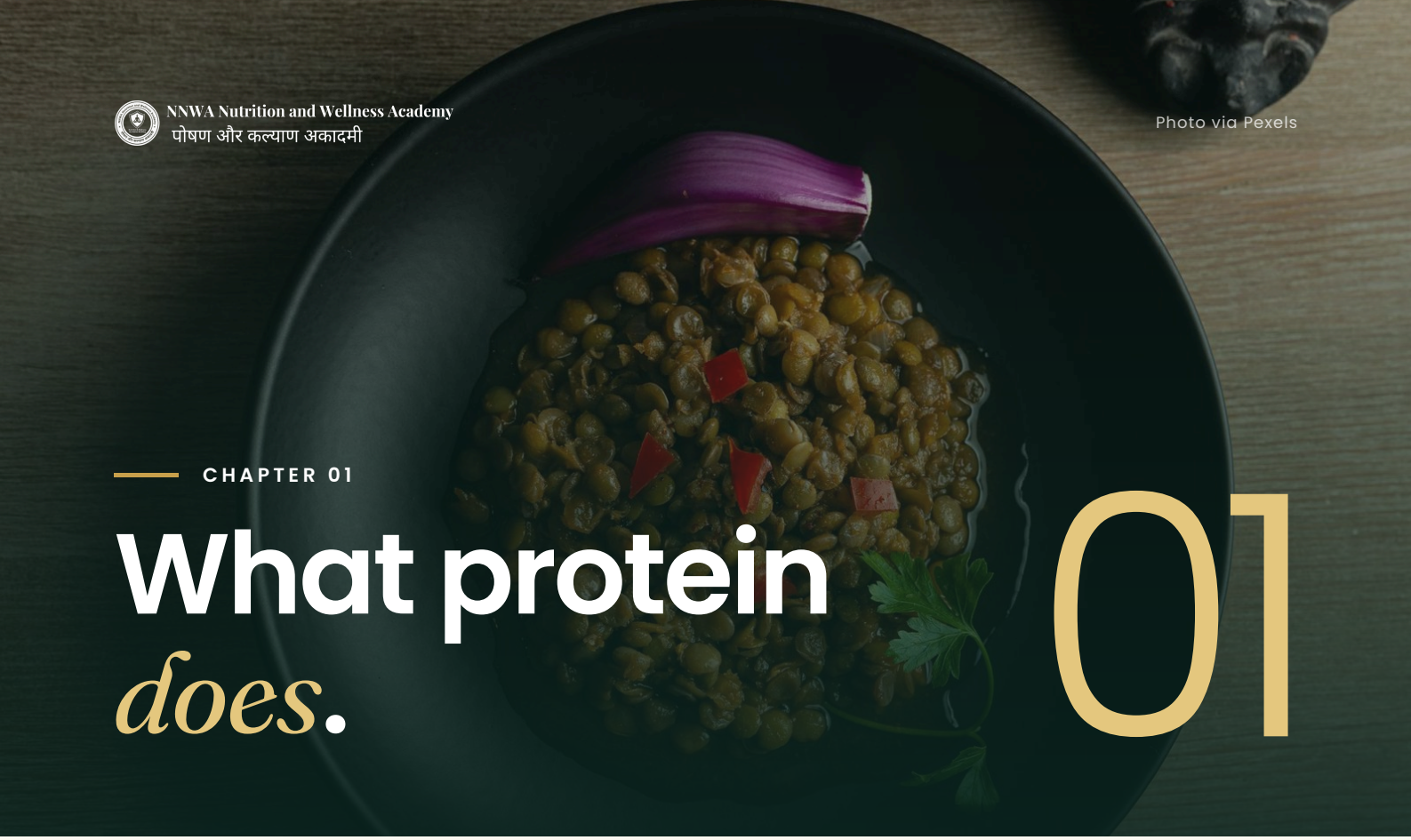


Photo via Pexels

CHAPTER 01

What protein *does*.

01

Protein is made of building blocks called amino acids. Your body uses about 20 of them. It can make some on its own; the rest, called essential amino acids, have to come from food every day.

9

essential amino acids that adults must get from food

01 Builds and repairs
Muscle, skin, hair, nails and organs are rebuilt from protein all the time.

02 Runs reactions
Enzymes, which drive digestion and metabolism, are proteins.

03 Sends signals
Some hormones, such as insulin, are proteins.

04 Defends
Antibodies that help fight infection are proteins.

05 Carries
Haemoglobin, the protein in red blood cells, carries oxygen.

06 Gives energy
About 4 kcal per gram, though the body prefers carbohydrate and fat for fuel.

WHY DAILY MATTERS

Unlike fat, the body keeps no dedicated store of protein to draw on. Body proteins are broken down and rebuilt every day, so a steady supply from meals keeps that cycle running.

HOW MUCH YOU NEED

One number to *remember*.

0.83 g/kg

grams of protein per kilogram of body weight, per day. This is the Recommended Dietary Allowance (RDA) for healthy Indian adults set by ICMR-NIN.

THE QUICK MATHS

Your weight in kg × 0.83 = your daily *grams*

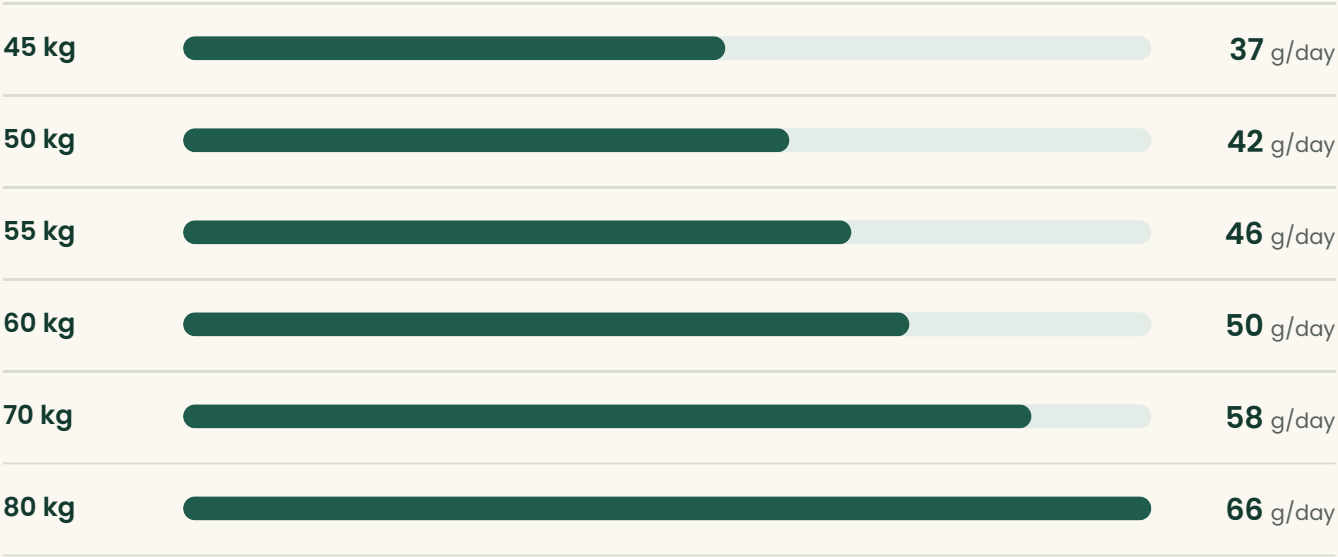
At 60 kg
≈ 50 g

Source: ICMR-NIN, Nutrient Requirements for Indians, 2020. An RDA is set to cover the needs of nearly all healthy people in a group, so many people need slightly less. General information only, not medical advice.

FIND YOUR NUMBER

Your weight, your *grams*.

Daily protein at 0.83 g per kg for healthy adults, rounded to the nearest gram.



WHEN THE NUMBER CHANGES

Pregnancy and breastfeeding	Needs go up. ICMR-NIN sets separate, higher amounts for each stage.
Children and teenagers	Needs are set by age and growth stage, not by the adult figure.
Older adults	Muscle loss with age makes protein important; the right amount is best set individually.
Kidney or liver disease	Some conditions need less protein, not more. Follow your doctor's advice.
Heavy training	Athletes may need more, within limits. See the myths on page 10.

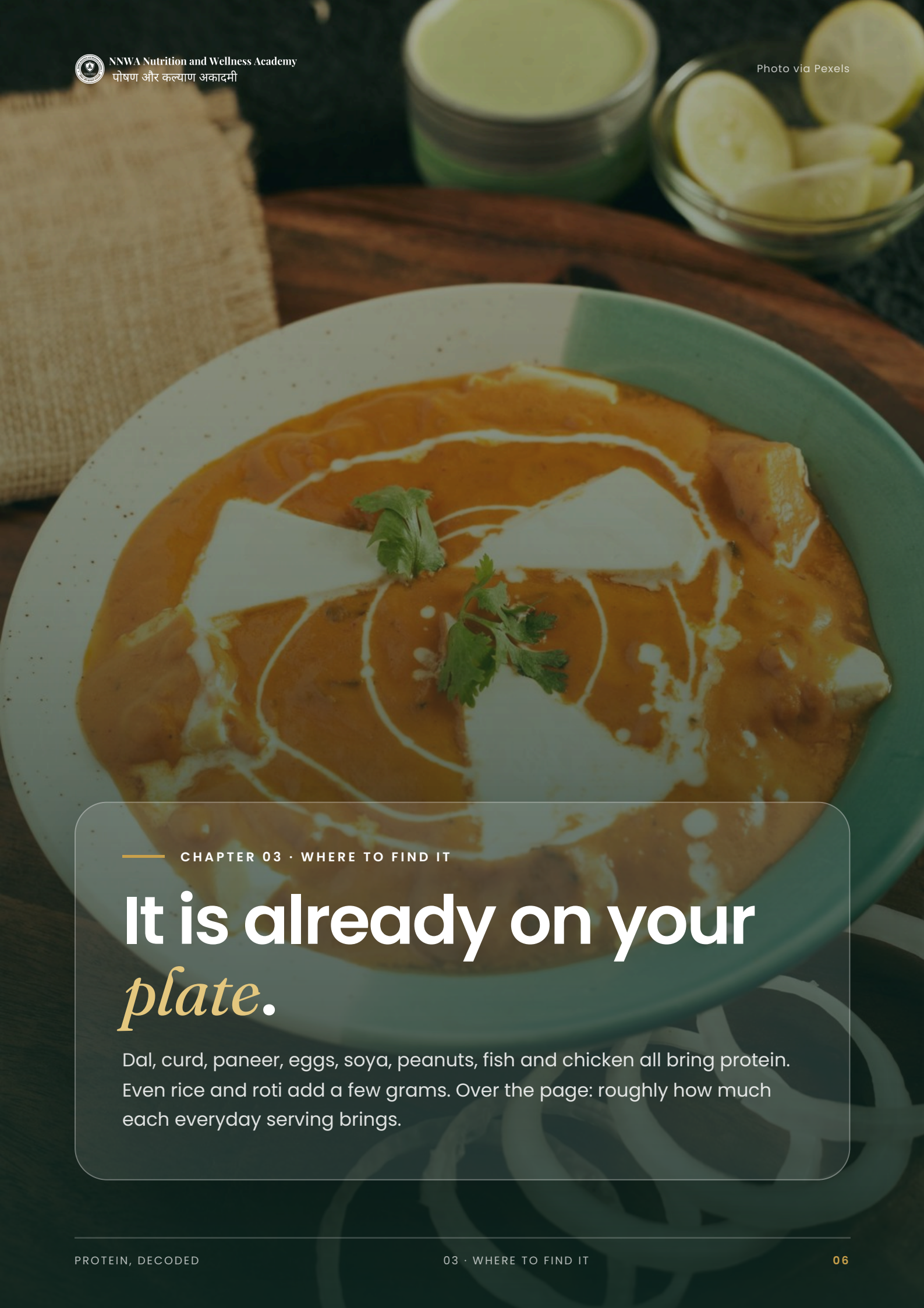


Photo via Pexels

CHAPTER 03 · WHERE TO FIND IT

It is already on your *plate*.

Dal, curd, paneer, eggs, soya, peanuts, fish and chicken all bring protein. Even rice and roti add a few grams. Over the page: roughly how much each everyday serving brings.

PROTEIN PER SERVING

Roughly how much, per *serving*.

PLANT FOODS

Soya chunks	25 g dry	**13 g**
Cooked dal	1 katori, from 30 g raw	**7 g**
Peanuts	30 g, a small handful	**7 g**
Roasted chana	30 g	**6 g**
Roti	1 medium, 30 g atta	**3 g**
Cooked rice	1 cup, from 50 g raw	**3.5 g**

DAIRY

Paneer	50 g	**9 g**
Milk	1 glass, 200 ml	**6 g**
Curd	1 katori, 150 g	**5 g**

EGGS, FISH AND MEAT

Chicken	100 g raw	**21 g**
Fish	100 g raw	**19 g**
Egg	1 large	**6 g**

Approximate, rounded values calculated from per-100 g figures in ICMR-NIN's Indian Food Composition Tables (IFCT 2017). Actual amounts vary with variety, recipe and portion. Bars are scaled to 13 g; chicken and fish run past the scale.

PAIR IT UP

Better *together*.



Cereals such as rice and wheat are low in the essential amino acid lysine. Pulses are low in methionine. Eaten in the same meal, each fills the other's gap, which is why so many Indian staples already pair them.

01
Dal and chawal
The everyday classic: rice with any dal.

02
Rajma or chole with rice
Kidney beans or chickpeas, with rice or roti.

03
Idli or dosa with sambar
Rice and urad in the batter, toor dal in the sambar.

04
Khichdi
Rice and moong dal cooked together in one pot.

05
Roti with chana or dal
Wheat and pulses, the North Indian everyday pair.

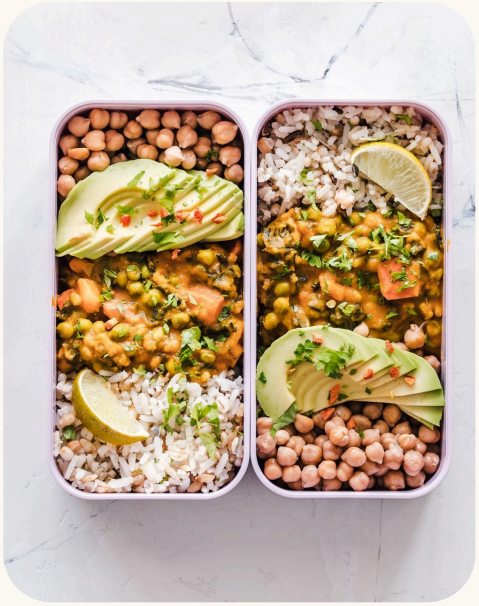
06
Poha with peanuts
Flattened rice with peanuts, which are a legume.

Milk, curd, paneer, eggs, fish, meat and soya provide all nine essential amino acids on their own. Adding a little of any of them to a cereal meal also lifts its protein quality. Photo via Pexels.

SPREAD IT ACROSS YOUR DAY

Little and *often*.

A protein food at every meal is easier than one big serving, and it fits how most Indian households already eat. Here is one illustrative day for a 60 kg adult.



BREAKFAST	**A glass of milk with an egg or a handful of roasted chana** Swap in curd, besan chilla or a sprouts bowl.	12 g
LUNCH	**Two rotis, a katori of dal, a katori of curd, sabzi** Cereal and pulse in the same meal.	18 g
SNACK	**A small handful of peanuts or sprouts chaat** Easy to carry; watch the portion size.	7 g
DINNER	**Rice with rajma, or 50 g paneer with sabzi** Fish or chicken works here too.	13 g

DAY TOTAL, APPROXIMATELY **50 g**

Illustrative only, not a meal plan. Values are approximate, from the serving table on page 7. Photo via Pexels.

MYTHS, CHECKED

Five things people *repeat*.

01

~~MORE PROTEIN ALWAYS MEANS MORE MUSCLE~~

Training builds muscle. Protein supports it, up to a point.

Muscle grows with resistance exercise plus enough protein and energy. ICMR-NIN's 2024 guidelines note that intakes above about 1.6 g/kg a day add no further training gains.

02

~~VEGETARIANS CANNOT GET ENOUGH PROTEIN~~

A varied vegetarian diet can meet the RDA.

Pulses with cereals, plus dairy, soya, nuts and seeds, cover the needs of most healthy adults.

03

~~PLANT PROTEIN DOES NOT COUNT~~

It counts. It just works best in combination.

Single plant foods may be low in one amino acid. Pairing cereals and pulses fills the gap (page 8).

04

~~YOU NEED A SHAKE AFTER EVERY WORKOUT~~

For most people, regular meals do the job.

Indian dietary guidance advises against protein supplements for building muscle (page 11).

05

~~PROTEIN IS ONLY FOR GYM GOERS~~

Everyone needs it, every day.

Repair, enzymes, immunity and oxygen transport all depend on protein, whatever your activity level.



Photo via Pexels

CHAPTER 07 · SUPPLEMENTS, CAREFULLY

Start with *food*.

ICMR-NIN's Dietary Guidelines for Indians (2024) advise against protein supplements for building muscle and recommend meeting needs through food combinations instead. They warn that large amounts of protein powder taken over long periods may be linked to bone mineral loss and kidney damage.

IF YOU ARE STILL CONSIDERING ONE

01 Speak to a qualified healthcare professional first, especially if you have any medical condition.

02 Check for an FSSAI licence number on the pack, and buy from a traceable seller.

03 Read the protein per serving, the serving size, and the added sugar and ingredients list.

04 Be wary of labels promising quick muscle, weight loss or other health results.

YOUR DAILY CHECKLIST

Eight small *habits*.

My weight _____ kg × 0.83 = _____ g protein a day

- [ ] **I know my number**
  Worked out above, and checked with a professional if any of page 5 applies to me. — ONCE
- [ ] **Protein at breakfast**
  Milk, curd, egg, chana, sprouts or besan. — DAILY
- [ ] **Protein at lunch**
  Dal, rajma, chole, paneer, fish or chicken. — DAILY
- [ ] **Protein at dinner**
  Another serving, not the whole day's worth. — DAILY
- [ ] **A cereal and pulse together**
  Dal-chawal, khichdi, idli-sambar or roti-chana. — DAILY
- [ ] **A dairy or soya serving**
  Curd, milk, paneer, tofu or soya chunks. — DAILY
- [ ] **Nuts or seeds as a snack**
  A small handful, not the whole packet. — MOST DAYS
- [ ] **Label check before any supplement**
  FSSAI number, serving size, added sugar, professional advice. — EVERY TIME

SOURCES AND NOTES

Where the numbers *come* from.

01 **ICMR-National Institute of Nutrition. Nutrient Requirements for Indians. 2020.**
Adult protein RDA of 0.83 g/kg body weight per day; separate requirements for pregnancy, lactation, children and adolescents. Used on pages 4, 5 and 9.

02 **ICMR-National Institute of Nutrition. Dietary Guidelines for Indians. 2024.**
Guidance on protein quality through food combinations, advice against protein supplements for muscle building, and the note that intakes above about 1.6 g/kg add no further training gains. Used on pages 8, 10 and 11.

03 **ICMR-National Institute of Nutrition. Indian Food Composition Tables (IFCT). 2017.**
Per-100 g protein values used to calculate the approximate serving figures on pages 7 and 9. Serving figures are NNWA's rounded calculations, not values printed in IFCT.

04 **General physiology**
Amino acids, the nine essential amino acids, protein's roles and its energy value of about 4 kcal per gram are standard textbook nutrition science (page 3).

05 **Photographs**
Free stock photos from Pexels (Pexels licence): IDs 11369845 (cover), 35763746 (p. 3), 12737805 (p. 6), 6187598 (p. 8), 1640769 (p. 9), 4491279 (p. 11). No person pictured or photographer endorses NNWA.

IMPORTANT

This guide is general educational information for healthy adults. It is not medical advice and does not diagnose, treat or prevent any condition. Individual needs vary. If you are pregnant or breastfeeding, under 18, an older adult, training heavily, or living with kidney, liver or any other medical condition, consult a qualified healthcare professional before changing your diet. NNWA (Nutrition & Wellness Academy) is an education provider.

KEEP LEARNING

Understand the science on your *plate*.

NNWA (Nutrition & Wellness Academy) is a 100% online academy for nutrition, wellness and fitness certification. This guide is one of a series of short, sourced nutrition basics.

EXPLORE PROGRAMMES

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